

Sustainable Forest Management

Science

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Sustainable Forest Management (A07691)

Short Title: Sustainable Forest Management
Faculty Science and Computing
Department Science
Cluster Forestry
Author ABOOTH
Credits 5

Level: Intermediate

Description of Module / Aims

This module aims at providing students with the foundations for the professional practice of sustainable forest management at a forest operational level in Ireland. This module will facilitate students in gaining knowledge of the International and national framework and implementation mechanisms for practicing sustainable forest management (SFM). Students will develop key professional skills for use in the application of SFM practices.

Programmes

FORS-0010 BSc in Forestry (WD_SFORS_D)

3 / 5 / M

Indicative Content

- International and Irish forest policy, strategies and standards
- Forest management objectives; identification of needs; historical developments; application of silvicultural systems; forest organisation
- Rotations; sustained yield; normal forest; felling decision; thinning systems
- Sustainable development; environmental, social and economic objectives; sustainable forest management principles and criteria; key indicators of SFM in Ireland
- Quality management in forestry; forest certification procedure, chain of custody, labelling
- Forest legislation in Ireland

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Relate international and Irish forest policy framework and legal requirements to managing a sustainable forest resource.
2. Examine the principles of sustainable forest management and their application in Irish forestry, control the manipulation of the forest through thinning spacing and stocking and be able to formulate clear and concise management objectives.
3. Apply knowledge of the certification systems applied to the forest sector and demonstrate the ability to carry out a basis certification assessment.
4. Employ the principal requirements for implementing a silvicultural system from even-aged clearcut plantation forestry to single tree selection systems and be able to develop a management plan for the application of an alternative silvicultural system.

Learning and Teaching Methods

- Lectures: will be used to develop understanding of the key concepts of sustainable forest management, the international and Irish regulatory framework for forest practice and the development of multiple purpose forestry to meet economic, environmental and societal needs on a sustainable basis.

- Fieldwork: will be used to develop professional competence in on-site forest certification procedures, implementing alternative silvicultural systems and assessing the practice of sustainable forest management at operation level.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,
Continuous Assessment	50%	
Assignment	50%	3,4

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	12	
Independent Learning	87	

Essential Material

- Forest Service, Anon. *_Code of Best Forest Practice_*. Ireland: Forest Service, Dept. Of Agriculture, Fisheries & Food, 2000.
- Forest Service, Anon. *_Indicative Forestry Statement_*. Ireland: Forest Service, Dept. Of Agriculture, Fisheries & Food, 2008.

- Forest Service, Anon. *_Irish National Forest Standard_*. Ireland: Forest Service, Dept. Of Agriculture, Fisheries & Food, 2000.
- Higman, S. *_Sustainable Forestry_*. UK: Earthscan Publishers, 1995.
- Maguire, B. *_A review of legislation that impacts on Irish forestry_*. Ireland: COFORD, 2001.
- Mathews, J.D. *_Silvicultural Systems_*. UK: Oxford Science Publishers, 1991.

Supplementary Material

- Evans, J. *_The Forests Handbook, an Overview of Forest Science_*. UK: Blackwell Science, 2000.
- Ferguson, I.S. *_Sustainable forest management_*. UK: Oxford University Press, 1996.
- Hart, C. *_Alternative silvicultural systems to clear cutting in Britain : a review_*. UK: HMSO, 1995.
- Sayer, J. *_Earthscan Reader in Forestry and Development_*. UK: Earthscan, 1999.
- Upton, C. and S. Bass. *_The Forest Certification Handbook_*. UK: Earthscan Publishers, 2006.